14.1.1. Complex Number Addition using Class

Algorithm: Complex Number Addition using Class

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1. Start

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2. Define Class

- Create a class named Complex

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3. Define Method: initComplex()

- Input real and imaginary parts
- Store them as:
  - self.real
  - self.imag

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4. Define Method: sum(c1, c2)

- Compute sum of two complex numbers:
  - self.real = c1.real + c2.real
  - self.imag = c1.imag + c2.imag

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5. Define Method: display()

- If imaginary part ≥ 0:
  - Print → real + imag i
- Else:
  - Print → real - |imag| i

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6. Create Objects

- Create three objects:
  - c1 → first complex number

- c2 → second complex number
- c3 → result

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7. Input Values

- Call c1.initComplex() → take input
- Call c2.initComplex() → take input

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8. Compute Sum

- Call c3.sum(c1, c2)

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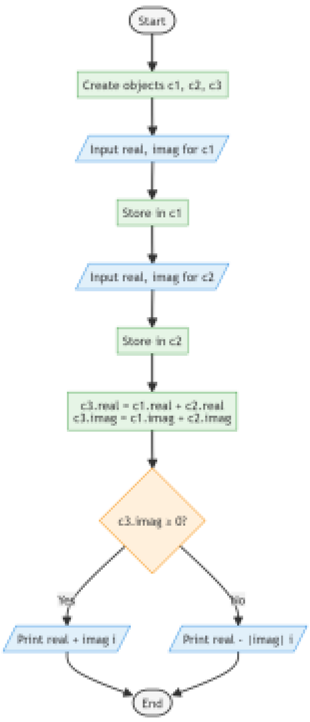
9. Display Result

- Call c3.display()

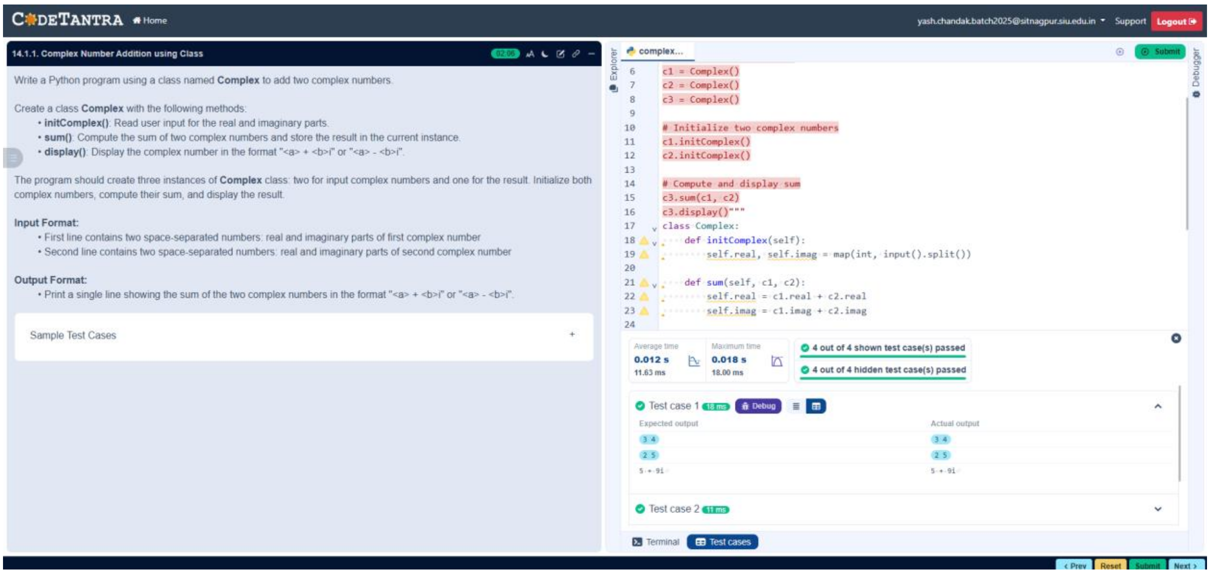
---

10. End



Flowchart :

Programme :



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14.1.1. Complex Number Addition using Class
Write a Python program using a class named Complex to add two complex numbers.
Create a class Complex with the following methods:
• initComplex(): Read user input for the real and imaginary parts.
• sum(): Compute the sum of two complex numbers and store the result in the current instance.
• display(): Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".
The program should create three instances of Complex class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.
Input Format:
• First line contains two space-separated numbers: real and imaginary parts of first complex number
• Second line contains two space-separated numbers: real and imaginary parts of second complex number
Output Format:
• Print a single line showing the sum of the two complex numbers in the format "<a> + <b>i" or "<a> - <b>i".
Sample Test Cases
complex...
Submit
c1 = Complex()
c2 = Complex()
c3 = Complex()
# Initialize two complex numbers
c1.initComplex()
c2.initComplex()
# Compute and display sum
c3.sum(c1, c2)
c3.display()"""
class Complex:
def initComplex(self):
self.real, self.imag = map(int, input().split())
def sum(self, c1, c2):
self.real = c1.real + c2.real
self.imag = c1.imag + c2.imag
Average time
0.012 s
11.63 ms
Maximum time
0.018 s
18.00 ms
4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
18 ms
Debug
Expected output
3 4
2 5
5 + 9i
Actual output
3 4
2 5
5 + 9i
Test case 2
11 ms
Terminal
Test cases
< Prev
Reset
Submit
Next >